Genómica Humana
Problem Set Genómica de Poblaciones

Fecha de entrega
21 de abril (Los problemas 1. – 5. 35 points total)
28 de abril (Ejercicio de reflexión/respuesta, 15 points)

Genetic variation
Problem 1.2. Using table 1.2 (Gilespie), calculate the frequency of the three alkaline phosphatase alleles in the English population.
(2 points)

Genetic drift
Problem 2.2. Write a simulation of genetic drift. Provide your code and resulting plot.
(5 points)

Problem 2.4 Graph Ht and Gt for 100 generations with Ho = 1 and population sizes of N=1, 10, 100, and 1,000,000.

(5 points)

Problem 2.10 If the human generation time is 20 years, what effective population size for humans is implied by the data?

(2 points)

Natural selection

Problem 3.1. In 1940, the frequency of the medionigra allele in the Oxford population was about p = 0.1. If the viabilities of the three genotypes were w11 = 0.9, w12 = 0.95, and w22 = 1, what would be the frequency of medionigra in the newborns of 1941?

(2 points)

Problem 3.2 *If the fitnesses of the genotypes* A_1A_1, A_1A_2 *and* A_2A_2, *are* w_{11} *=1.5,* w_{12} *=1.1, and* w_{22} *=1.0, respectively, what are the values of the selection coefficient and the heterozygous effect?*

(2 points)

Problem 3.5 *Write down Equation 3.2 for the special case* $h = 0.5$. *Find the allele frequencies in two successive generations* (p' and p'') *when the initial value of* $p = 0.1$ *and* $s = 0.1$. *Continue this for 200 generations and graph the result.*

(5 points)

Problem 3.7 *Graph $\Delta_s p$ versus p for an underdominant locus.*

(5 points)

Linkage disequilibrium
Problem 4.1. Derive the three equations for the frequencies of the A1B2, A2B1, and A2B2 gametes after a round of random mating.
(2 points)

Population subdivision

Problem 5.1 *Find p and F for a population in which the genotype frequencies of A_1A_1, A_1A_2, and A_2A_2 are 0.056, 0.288, and 0.656, respectively.*

(2 points)

Problem 5.10 *Calculate F_{ST} for the data in Table 1.3. Pretend that there are only two alleles by using the S alleles for A_1 and lumping the other alleles into A_2. First, assume that all of the subdivisions are the same size (c_i = 1/8). Next, assume that*

$$c_i = \begin{cases} 0.2 & i = 1,2,3,4 \\ 0.05 & i = 5,6,7,8. \end{cases}$$

Does the fact that F_{ST} depends on your assumption about the c_i make you nervous?

(3 points)

Problem 6.4. The following measurements of weights are from a single parent and its offspring and are expressed as deviations from the mean. What is the heritability of this trait? (2 points)

(-0.002, -0.391) (1.566,1.747) (0.542, -2.127) (-0.285, -1.623) (-1.519, -0.876) (-1.136, -0.705) (0.907, 0.458) (0.435, -0.287) (1.292, 0.153) (-0.640,0.711)

While some solutions are provided in Gilespie, the idea is that you work through the problem yourself to develop an understanding. If you simply copy the solution from the text, no credit will be given.
Aunque Gilespie ofrece algunas soluciones, la idea es que resuelvas el problema por tu cuenta para comprenderlo (mostrando su trabajo y proceso de pensamiento). Si simplemente copias la solución del texto, no se te dará ningún crédito.

Ejercicio de reflexión/respuesta

Escriba un documento de reflexión/respuesta de 1-2 páginas (max 2 páginas, inglés o español). Elija una de las siguientes indicaciones:

- ¿Qué es una población?

- ¿Cómo ha afectado mi visión de la genómica el aprender la historia de la genética de poblaciones? ¿Los científicos tienen una responsabilidad en la sociedad y por qué?

- ¿Cómo han dado forma las ideas sobre la raza y la superioridad racial a las ideas y conceptos de la genética de poblaciones?

- La selección natural como marco para estudiar el mundo natural

- Patrones de apareamiento en el mundo actual. ¿Vemos apareamiento aleatorio o no aleatorio? ¿Qué tipos de migraciones, mezclas y subdivisiones vemos?

- La consanguinidad en el mundo de hoy

- Puede elegir otro tema siempre que esté dentro del tema de la genómica de poblaciones o sobre la relación de la ciencia con la sociedad

Write a 1-2 page reflection/response document (max 2 pages, English or Spanish). Choose one of the following prompts:
- What is a population?
- How has learning the history of population genetics affected my view of genomics? Do scientists have a responsibility in society and why?
- How have ideas about race and racial superiority shaped the ideas and concepts of population genetics?
- Natural selection as a framework for studying the natural world
- Mating patterns in the modern world. Do we see random or non-random mating? What types of migrations, mixtures and subdivisions do we see?
- Consanguinity in today's world
- You can choose another topic as long as it is within the topic of population genomics or on the relationship of science with society

Por favor escribe a mashaal@ccg.unam.mx con cualquier duda o pregunta.

Please write to mashaal@ccg.unam.mx with any questions or concerns.